

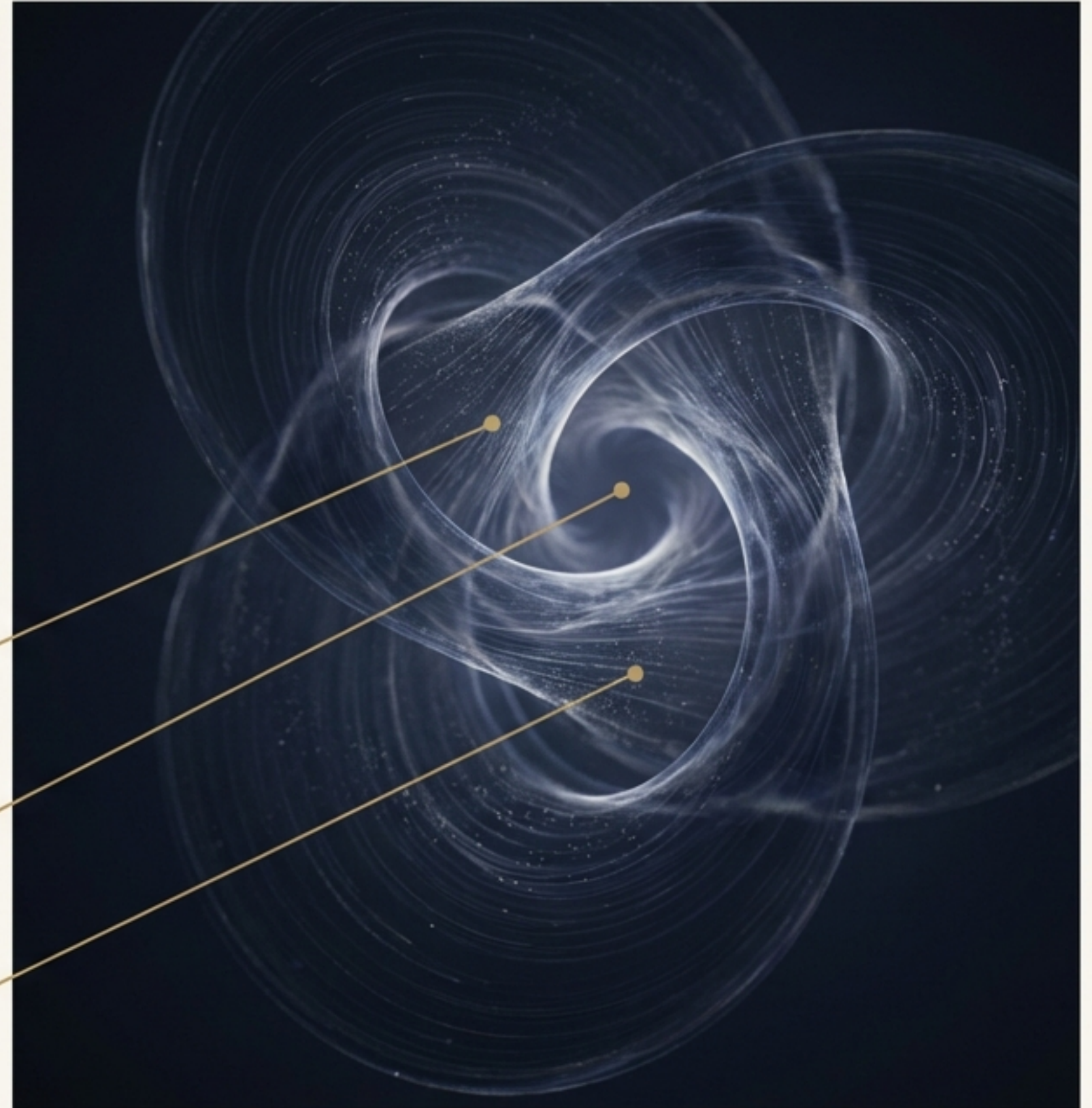
Time, Gravity, and Mass are Emergent Manifestations of a Single Hydrodynamic Substrate.

Modern physics rests on three pillars often treated as fundamental: the nature of Time, the mechanism of Gravity, and the origin of Mass. Swirl-String Theory (SST) proposes these are not fundamental. Instead, they emerge from the dynamics of a single, frictionless, incompressible superfluid condensate.

Time: is the local counting of vortex events.

Gravity: is the gradient in the density of these events.

Mass: is the energy trapped within topological defects (knots).



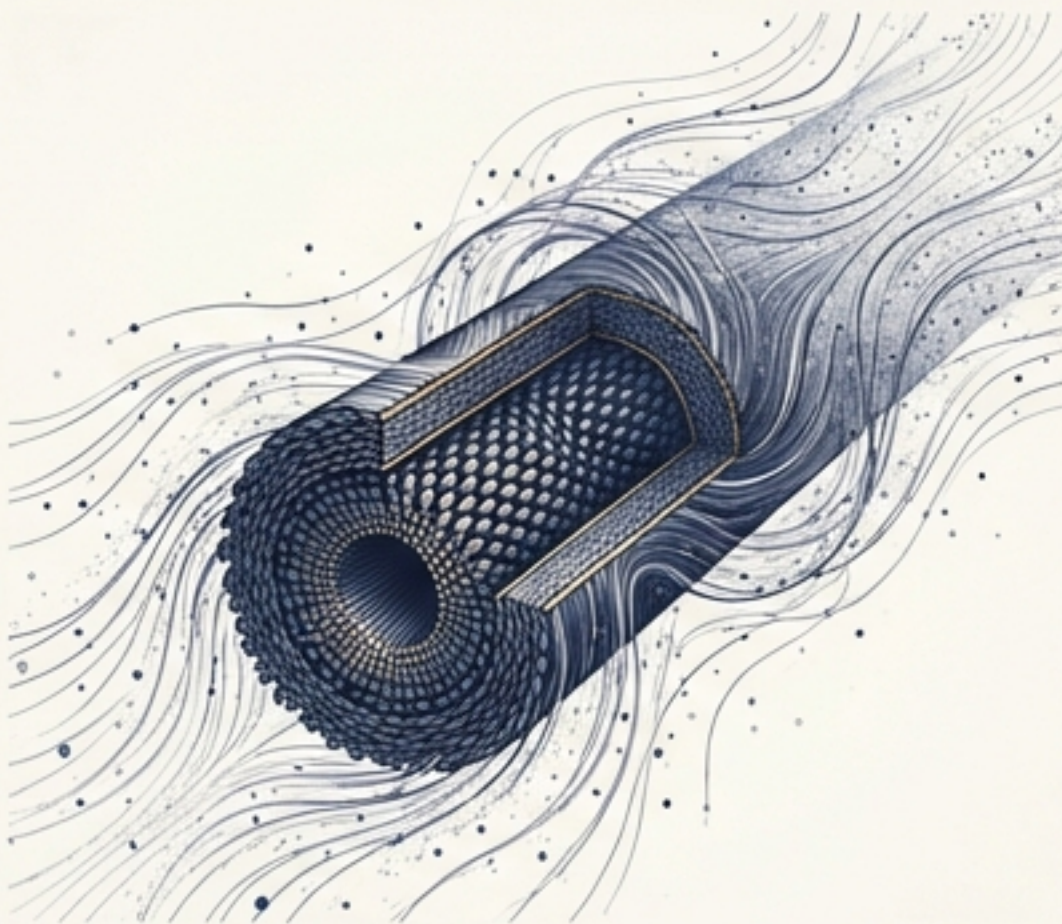
The Theory is Built on Three Canonical Axioms.



Axiom I: Swirl-Time & Foliation

Time is not a universal parameter t , but a local physical field $\chi(x)$. The local tick-rate is governed by the tangential swirl velocity v .

$$dt(x) = \mathcal{S}_o(x) dt_\infty = \sqrt{1 - \frac{|v|^2}{c^2}} dt_\infty$$



Axiom II: Incompressible Superfluid Vacuum

The universe is a perfect, inviscid fluid. It has two distinct densities:

- Vacuum Density: $\rho_f \approx 7.0 \times 10^{-7} \text{ kg m}^{-3}$
- Core Density (inside vortex filaments):
 $\rho_{\text{core}} \approx 3.89 \times 10^{18} \text{ kg m}^{-3}$



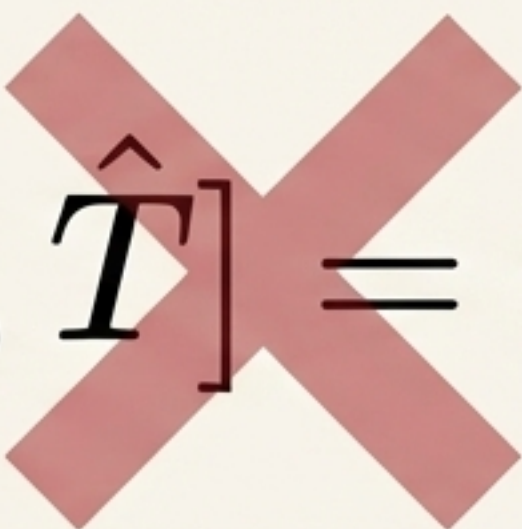
Axiom III: Topological Matter (Rosetta Rule)

Stable particles are closed, knotted vortex filaments. Conserved quantum numbers (charge, spin) correspond to topological invariants (linking number, twist, knot type).

To Ground This Vision, We First Solve a Foundational Problem: The Time-of-Arrival.

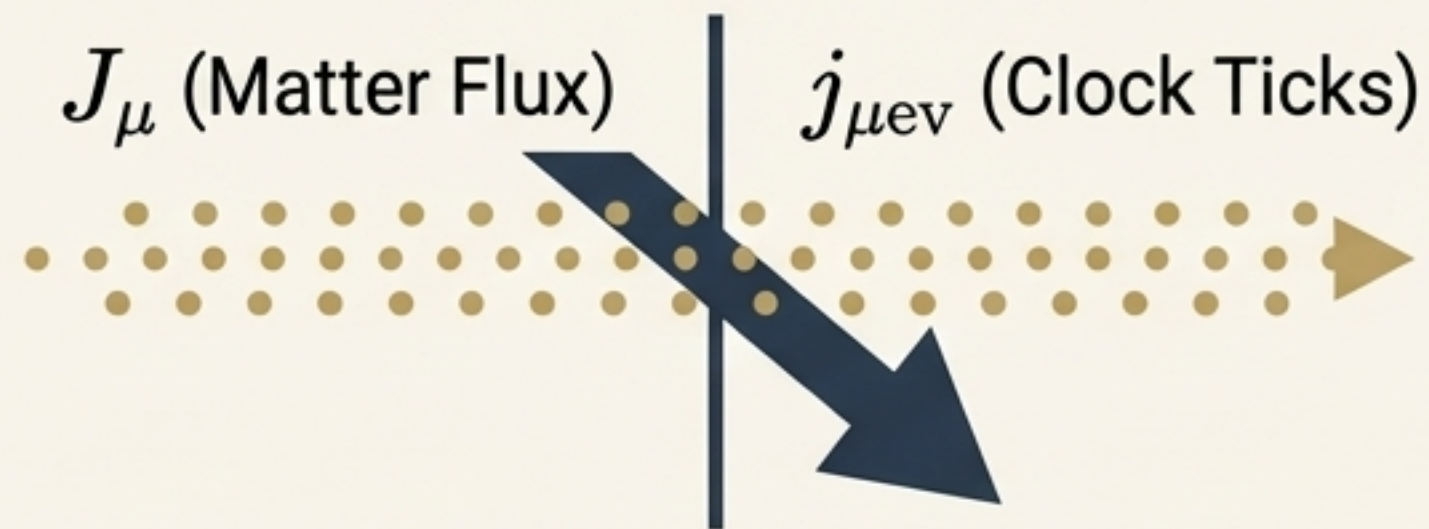
Quantum theory has long struggled with a consistent definition of time. Pauli's argument obstructs a self-adjoint time operator conjugate to a bounded Hamiltonian. Existing TOA constructions are often detector-dependent, tying the observable to a chosen clock variable or preferred slicing. We propose a different path: defining TOA as a *relational field observable*. We will not postulate a time operator; we will construct a physical clock from conserved currents.

Standard Approach


$$[\hat{H}, \hat{T}] = i\hbar?$$

Pauli's Objection: Obstructed by a bounded Hamiltonian.

Relational Approach



Relational Field Observable: Binning one physical process against another.

A Physical Clock is Constructed from Two Conserved Currents

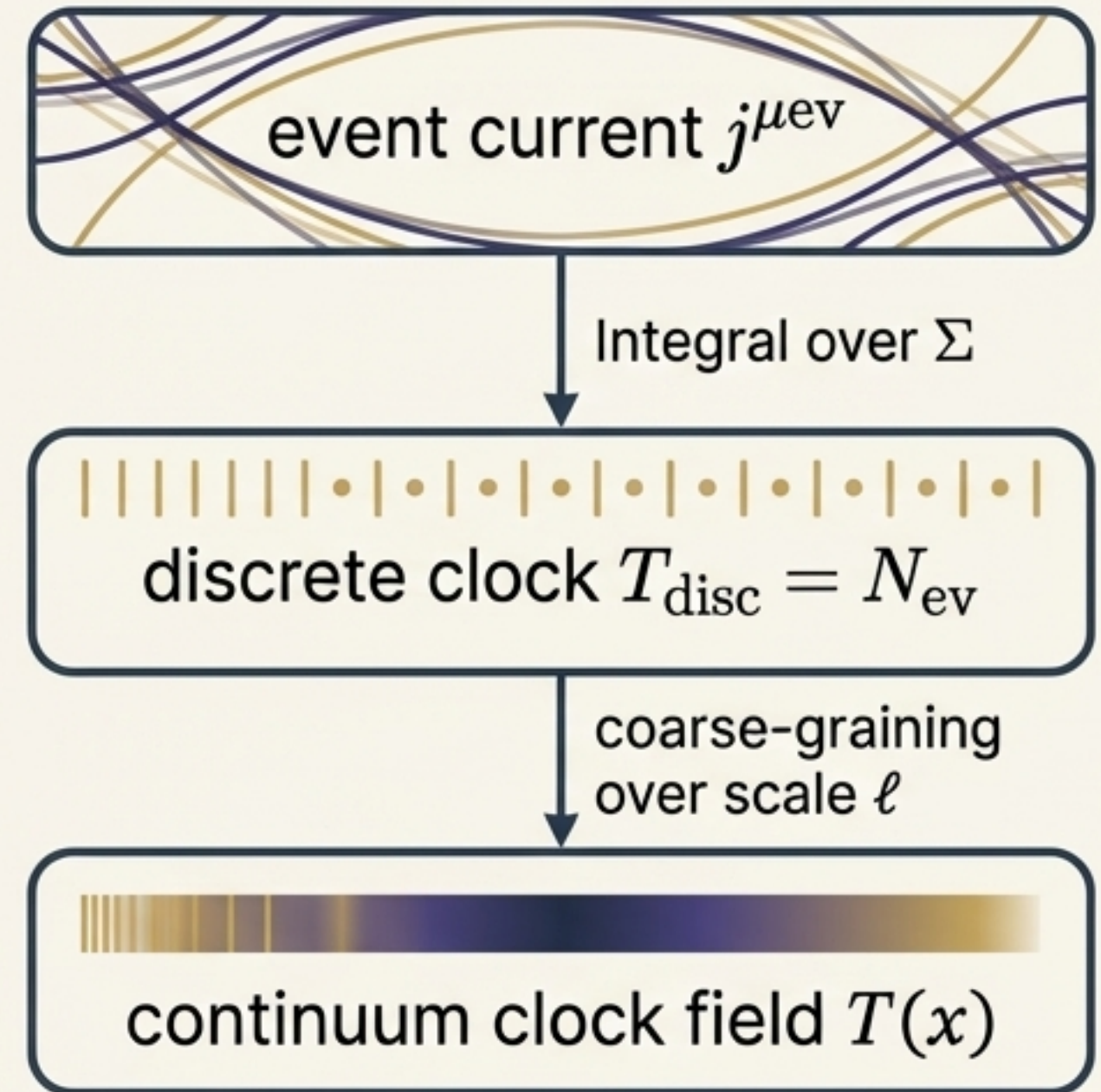
Core Ingredients

1. Matter Current (J^μ)

A conserved current ($\nabla_\mu J^\mu = 0$) that records the flux of particles through a detector's world-tube, Σ . This is *what* is being detected.

2. Event Current ($j^{\mu\text{ev}}$)

A second, approximately conserved current ($\nabla_\mu j^{\mu\text{ev}} \approx 0$) whose integral counts localized, robust physical events. This is *what defines the clock*.



The event current is the fundamental object. Discrete counts (UV) and continuum time (IR) are two descriptions of the same underlying physical structure.

Time-of-Arrival is the Detector Flux Binned by the Physical Clock Field

The relational TOA probability density is given by:

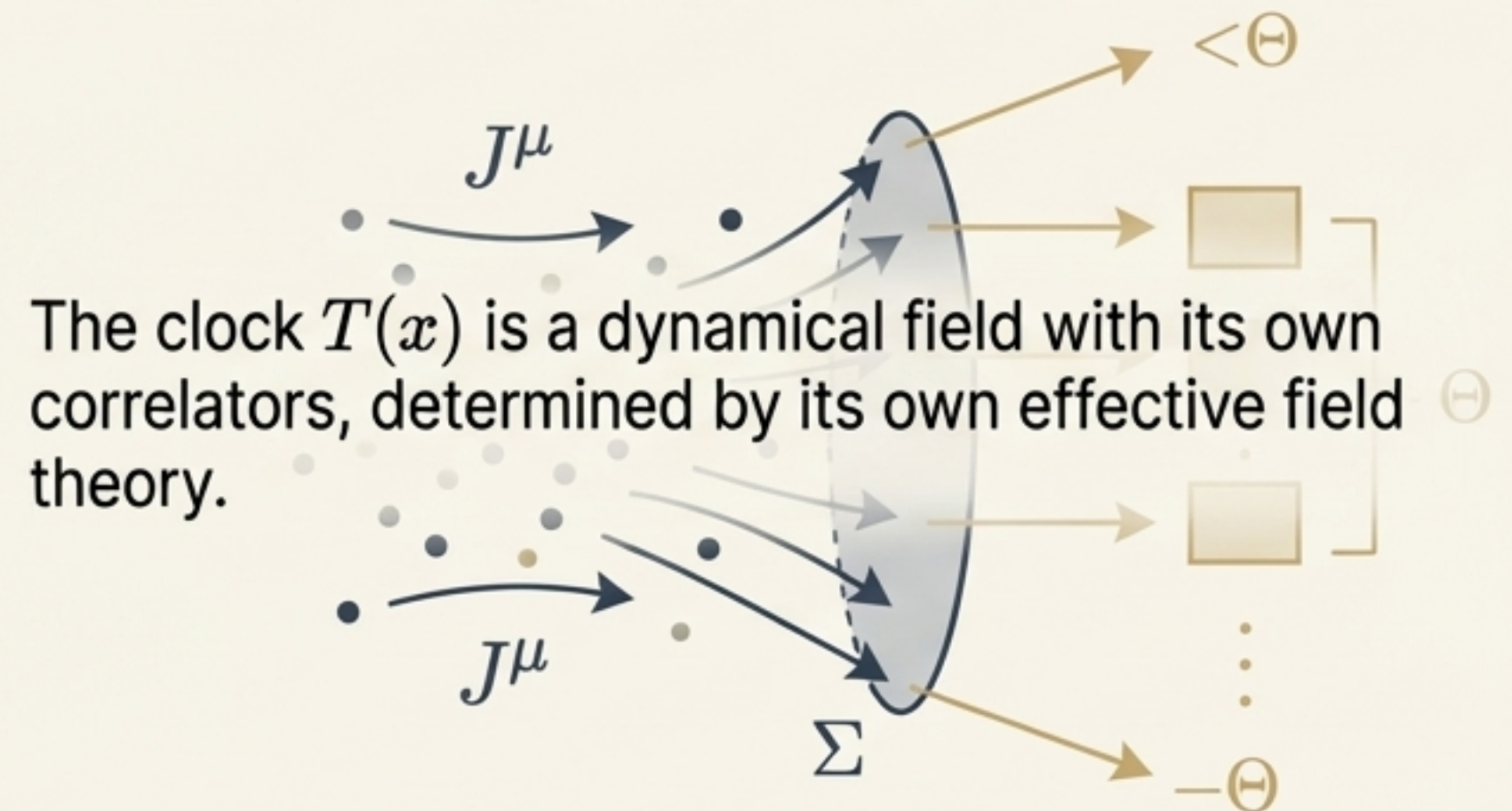
$$p(\Theta) \equiv \frac{1}{N} \int_{\Sigma} d^3\sigma \sqrt{|h|} F(\sigma) \delta(T_{\Sigma}(\sigma) - \Theta)$$

Where $F(\sigma) = n_{\mu} J^{\mu}$ is the matter flux and $T_{\Sigma}(\sigma)$ is the clock field evaluated on the detector.

Why This is Nontrivial

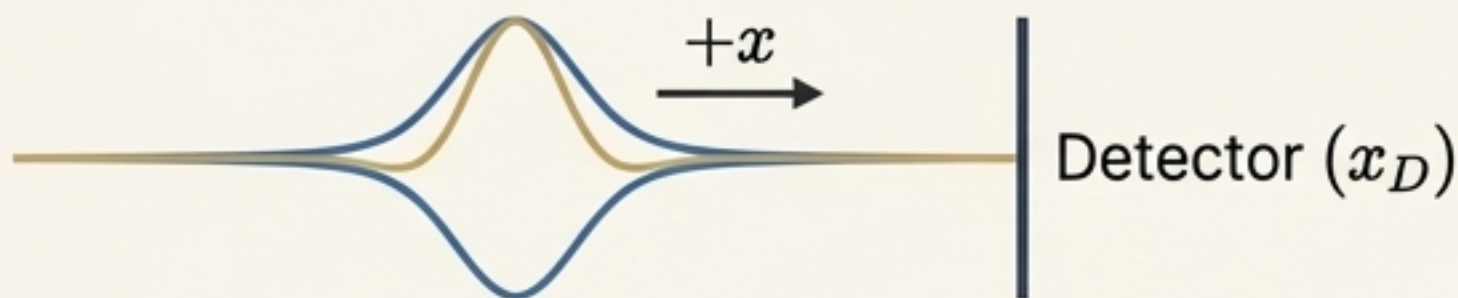
This is not a relabeling of an external time coordinate t .

The result is a covariant observable, independent of detector microphysics, that sidesteps Pauli-type obstructions from the start.



A 1+1D Calculation Reveals the Clock's Physical Impact

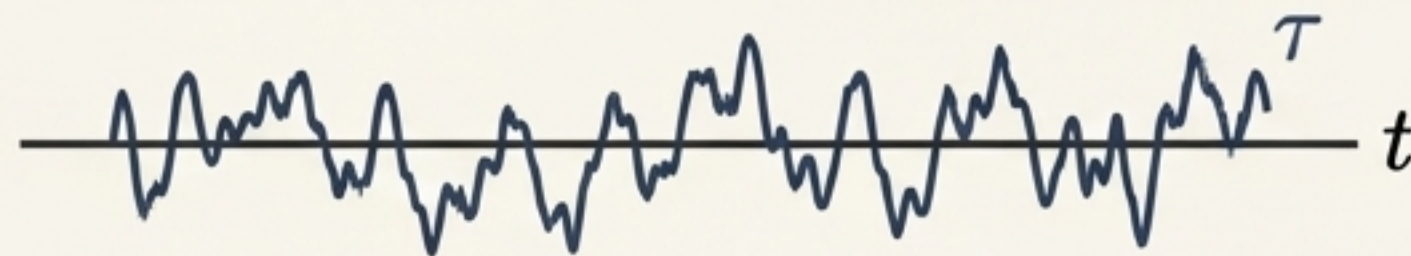
Setup: Matter Sector



A narrowband Klein-Gordon wavepacket propagating in the $+x$ direction. Its semiclassical flux at the detector x_D is an approximately Gaussian profile:

$$F(t) \approx F_0 \exp\left[-(t - t_{\text{cl}})^2 / 2\sigma^2 t\right]$$

Setup: Clock Sector



The continuum clock field is modeled as $T(t, x) = t + \tau(t, x)$, where τ represents fluctuations. We assume τ is a stationary Gaussian random variable with mean 0 and variance σ_τ^2 .

The Observable

The TOA density requires averaging over the clock fluctuations:

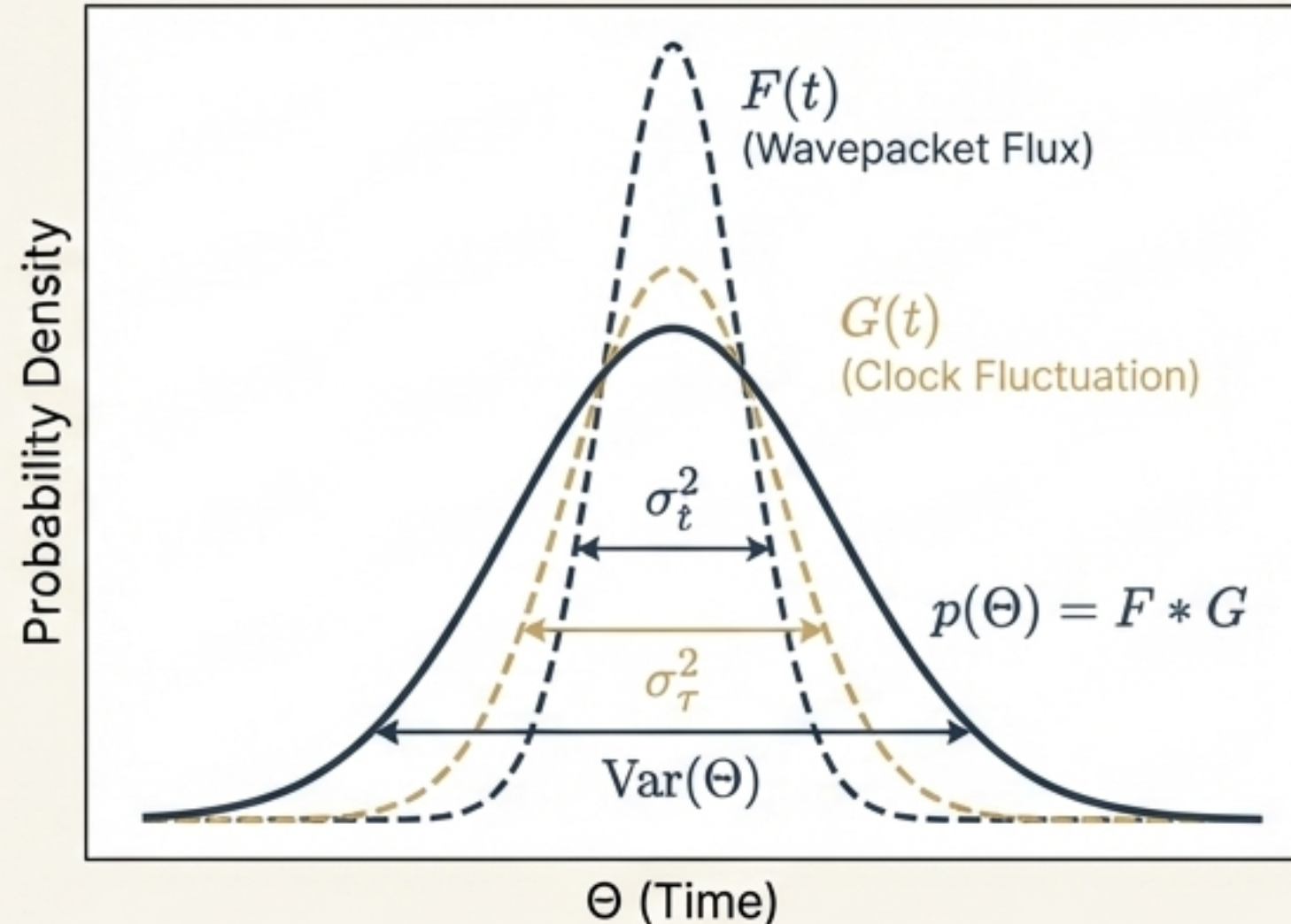
$$p(\Theta) = \frac{1}{N} \int dt F(t) \langle \delta(t + \tau(t, x_D) - \Theta) \rangle_\tau$$

The Result is a Convolution: The Clock's Fluctuations Broaden the Arrival Time

The Convolution Formula

The average over Gaussian clock fluctuations transforms the integral into a convolution:

$$p(\Theta) = \int dt F(t) * \frac{1}{\sqrt{2\pi\sigma_\tau^2}} \exp\left[-\frac{(\Theta - t)^2}{2\sigma_\tau^2}\right]$$



The Predictive Result

For a Gaussian flux profile $F(t)$, the resulting TOA distribution $p(\Theta)$ is also a Gaussian, but with an increased variance:

$$\text{Var}(\Theta) = \sigma_t^2 + \sigma_\tau^2$$

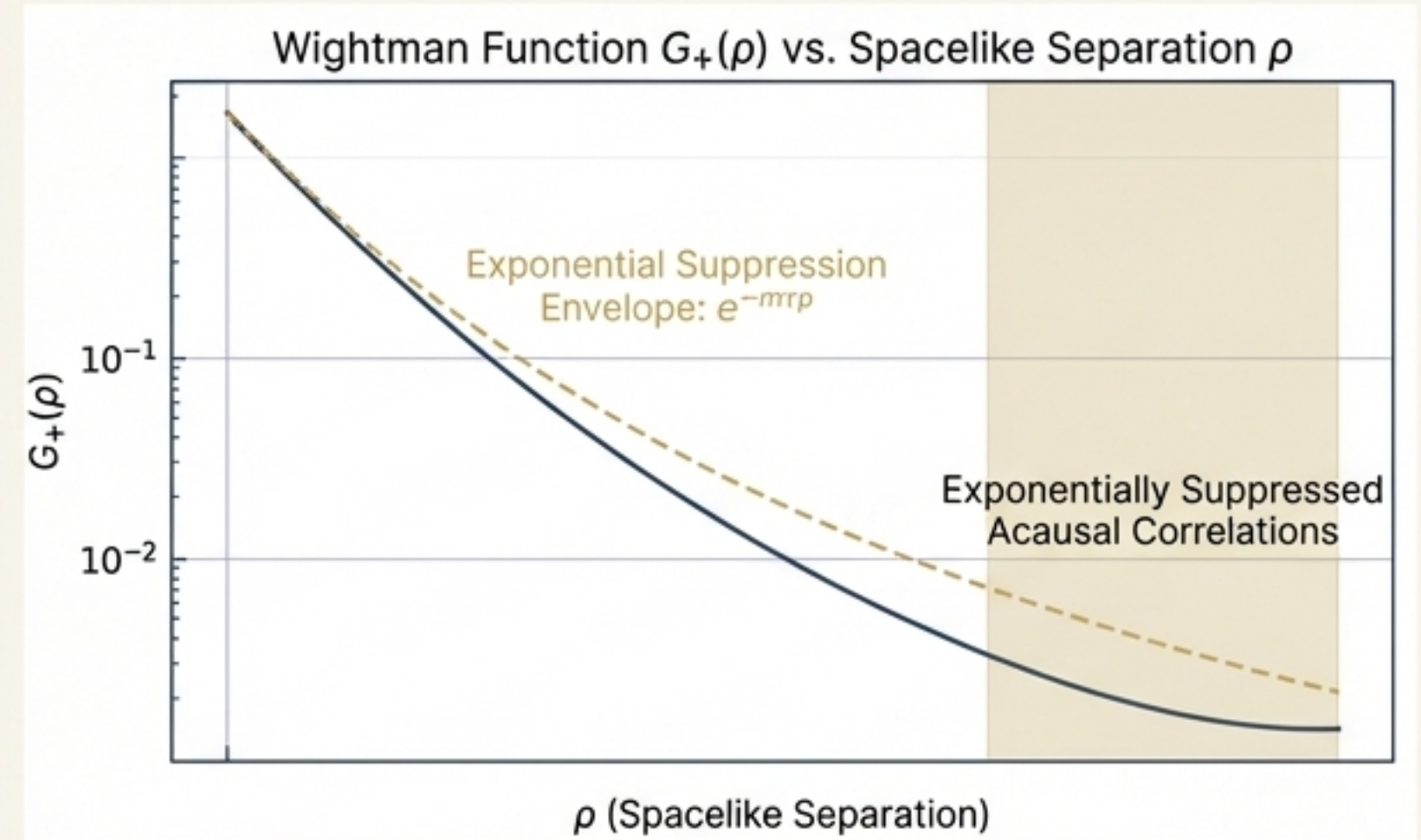
The Falsifiable Prediction: The clock's own quantum fluctuations (σ_τ^2) add a new, calculable source of variance to the arrival time. An experiment measuring $\text{Var}(\Theta)$ greater than the wavepacket's intrinsic spread σ_t^2 would provide direct evidence for this formalism.

The Clock's Dynamics Also Enforce a Causal Envelope

The clock field is described by a minimal stable quadratic EFT with a mass parameter μ_T . This parameter sets the correlation scale.

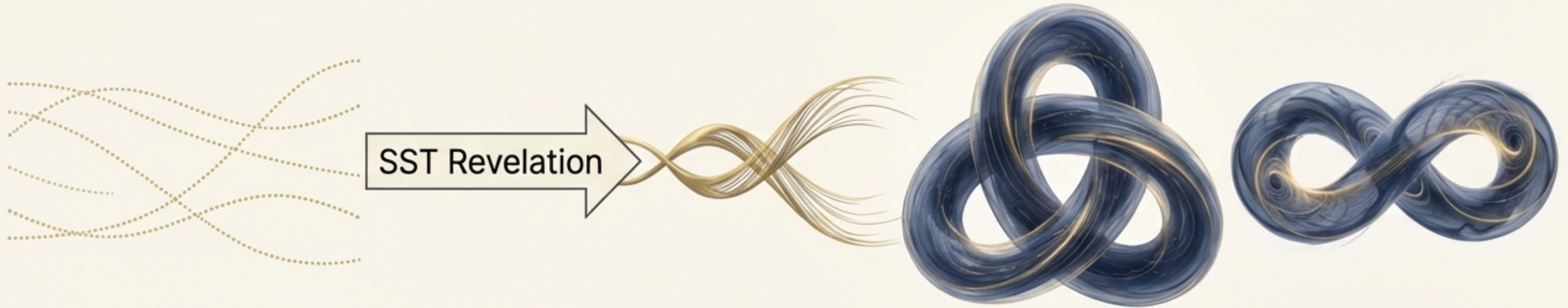
The Wightman function for this field shows that correlations at spacelike separations ρ are not strictly zero, but are exponentially suppressed.

$$G_+(\rho) \propto \frac{e^{-m_T \rho}}{\sqrt{\mu_T \rho}} \quad \text{for } \rho \gg \mu_T^{-1}$$



The Implication: Any contribution to the TOA that would require correlations across a spacelike interval is exponentially suppressed by the clock's mass scale μ_T . This provides a controlled bound on early-arrival probabilities without implying superluminal signaling.

What Physical Process Generates the ‘Event Current’ $j^{\mu\text{ev}}$?



The Answer from Swirl-String Theory:

- The abstract “events” of the TOA formalism are the fundamental particles of SST.
- The event current $j^{\mu\text{ev}}$ is the conserved topological current of knotted vortex filaments moving through the superfluid condensate.

$$j_{\mu\text{ev}}(x) \equiv \sum_k q_k \int d\tau \frac{dz^\mu_k}{d\tau} \delta^4(x - z_k(\tau))$$

The Physical Picture:

A “tick” of the clock is the physical event of a topological knot (an electron, a proton) crossing a surface. Time measurement is literally counting knot crossings.

The Clock Field and the Gravitational Field Are the Same Entity.

The Unified Field

- The continuum clock field $\chi(x)$ (the IR limit of $j_{\mu}^{\mu ev}$) is also the mediator of gravity.
- It represents the local “density of time,” or swirl pressure.
- Matter (vortex knots) acts as a source for this field. In the static, weak-field limit, it satisfies the Poisson equation:

$$\nabla^2 \chi(\mathbf{x}) = 4\pi G_{\text{eff}} \rho_{\text{matter}}(\mathbf{x})$$

Emergent Gravity

The Green's function solution is the familiar $1/r$ potential. Gravity emerges not from spacetime curvature, but as a hydrodynamic pressure gradient: matter migrates toward regions of slower time.

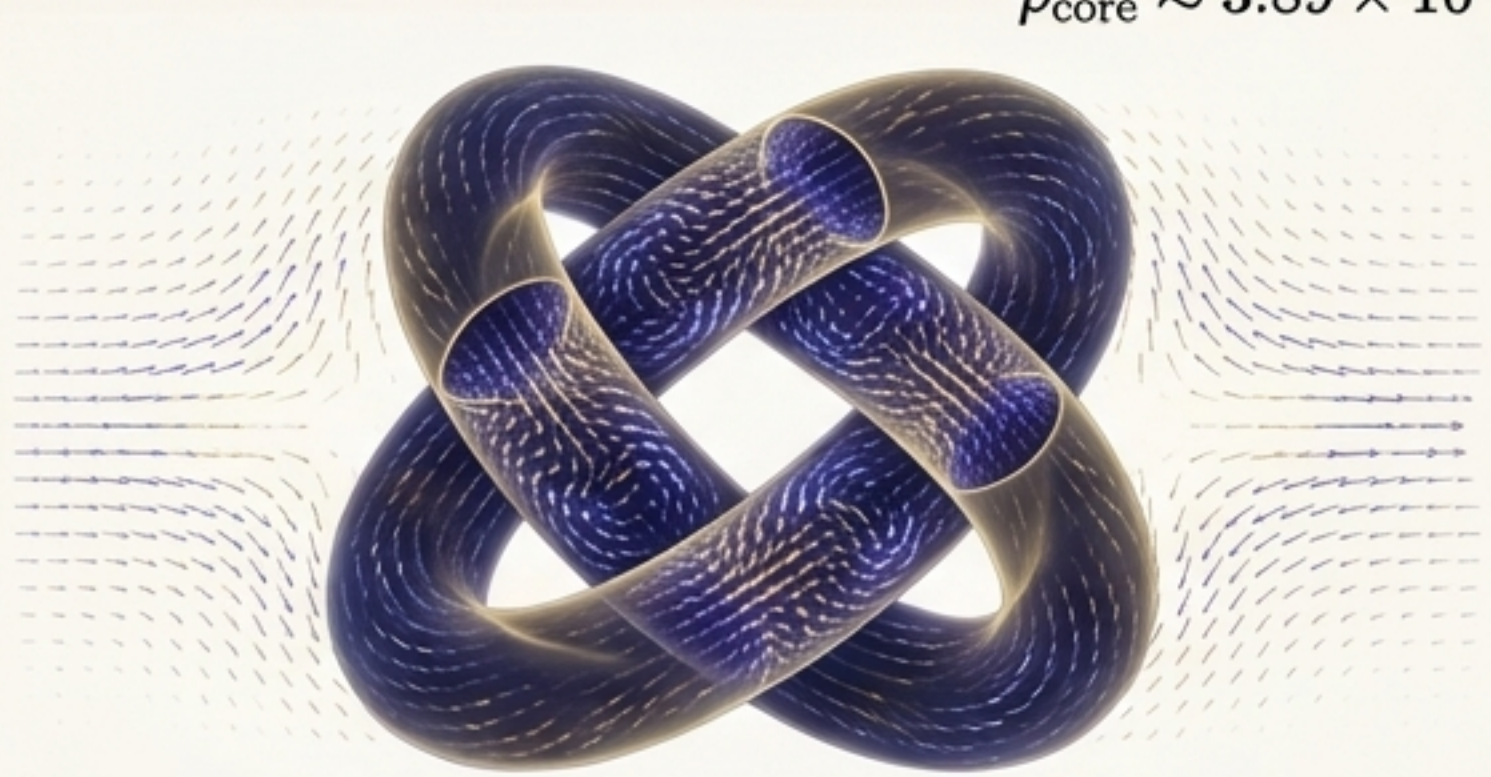


Invariant Mass is the Trapped Swirl Energy of a Topological Knot

The rest energy of a stable particle (a knot K) is the kinetic swirl energy localized in its hils core volume V_K :

$$M(K)c^2 \equiv \int_{V_K} \frac{1}{2} \rho_{\text{core}} |\mathbf{v}(\mathbf{r})|^2 d^3r$$

Note the use of the extremely high core density:
 $\rho_{\text{core}} \approx 3.89 \times 10^{18} \text{ kg m}^{-3}$.



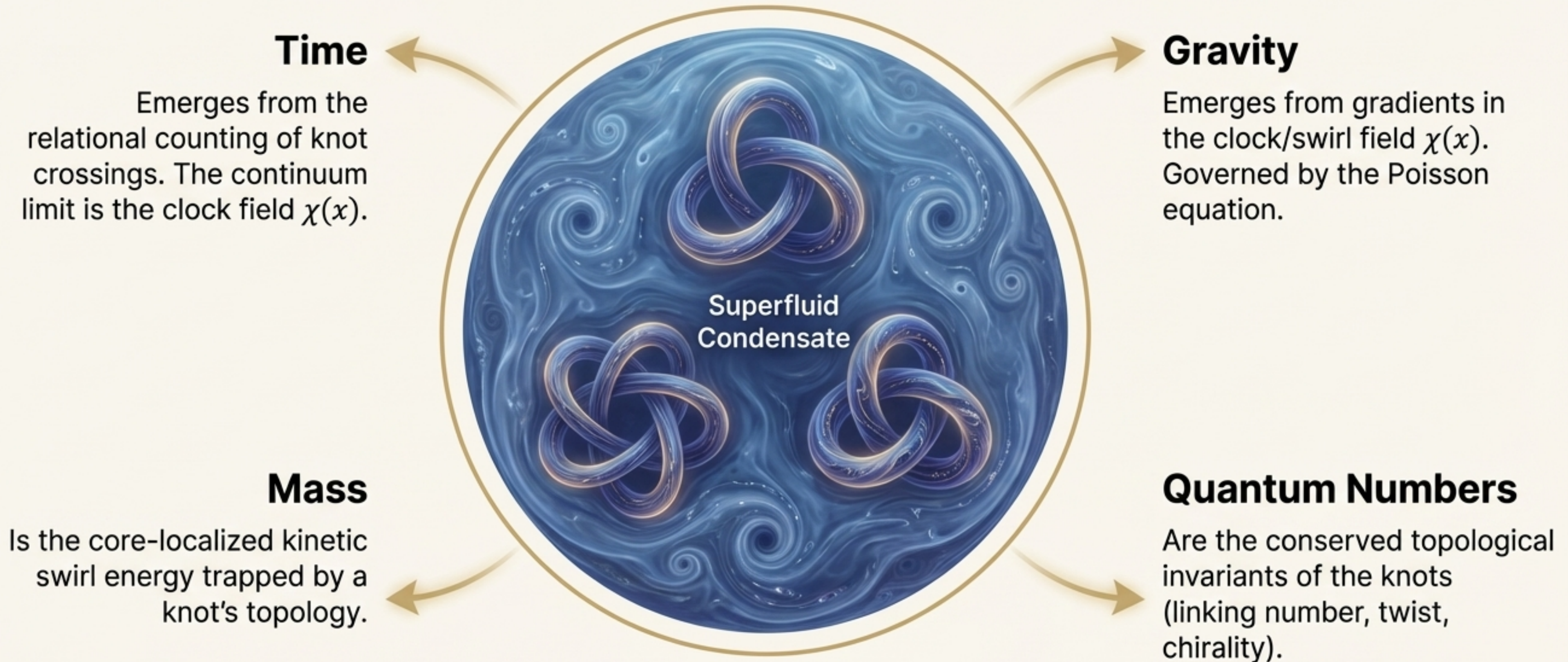
Discrete Mass Spectrum from Discrete Topology

- The velocity field $\mathbf{v}(\mathbf{r})$ is determined by the knot's topology.
- Stable knots fall into discrete complexity classes (e.g., trefoil vs. cinquefoil).
- This discreteness in topology leads to a discrete mass spectrum...

Knot Topology	Particle Analogy	Relative Mass
Simplest chiral knot	Electron	M_e
Complex composite knot	Proton	$\sim 1836 M_e$

...providing a structural explanation for the electron-proton mass hierarchy based on knot complexity.

The Hydrodynamic Canon: A Unified Ontology



The Theory Offers Further Falsifiable Predictions

Neutrino Coupling

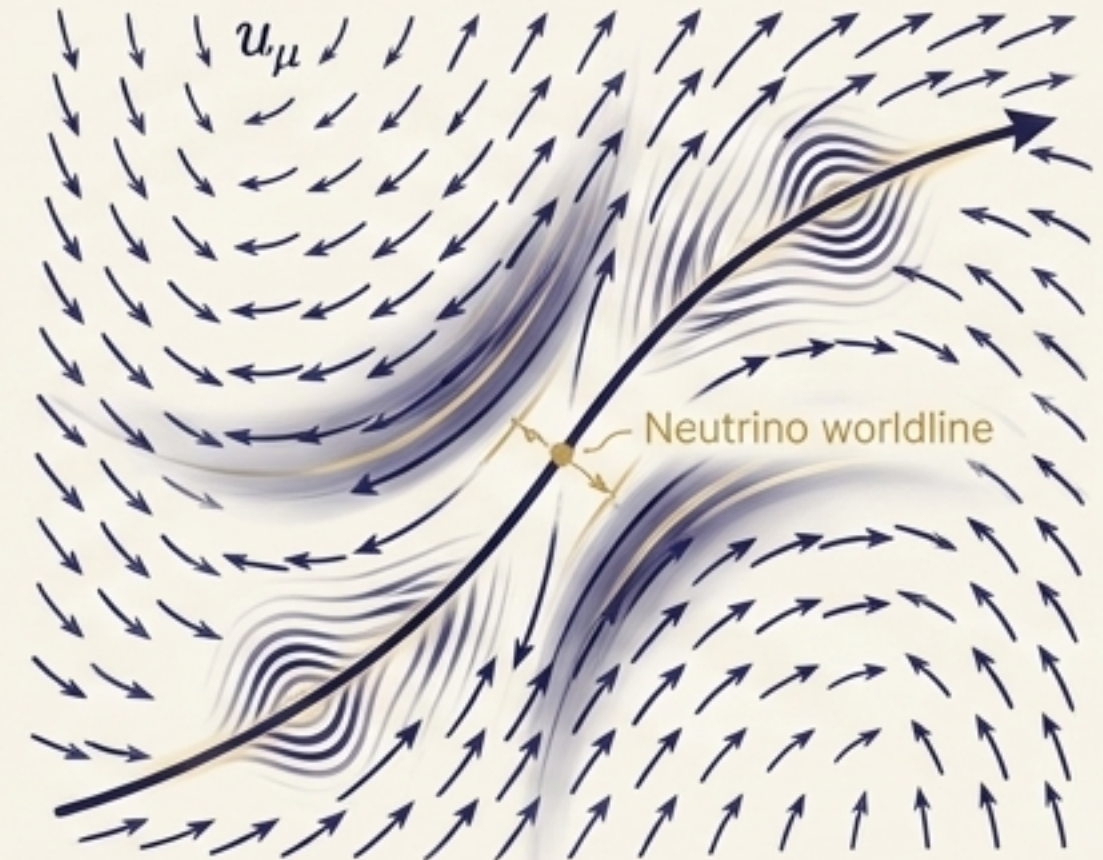
Beyond TOA Broadening: The unified structure leads to testable consequences in other domains.

Clock-Frame Effects on Neutrinos:

- The clock field $\chi(x)$ defines a physical, unit timelike vector field u^μ , representing the local rest frame of the condensate.
- SST predicts a minimal, chirality-selective coupling between neutrinos and this clock frame:

$$L_{\nu u} = \lambda_\nu u_\mu \bar{\nu} \gamma^\mu \gamma^5 \nu$$

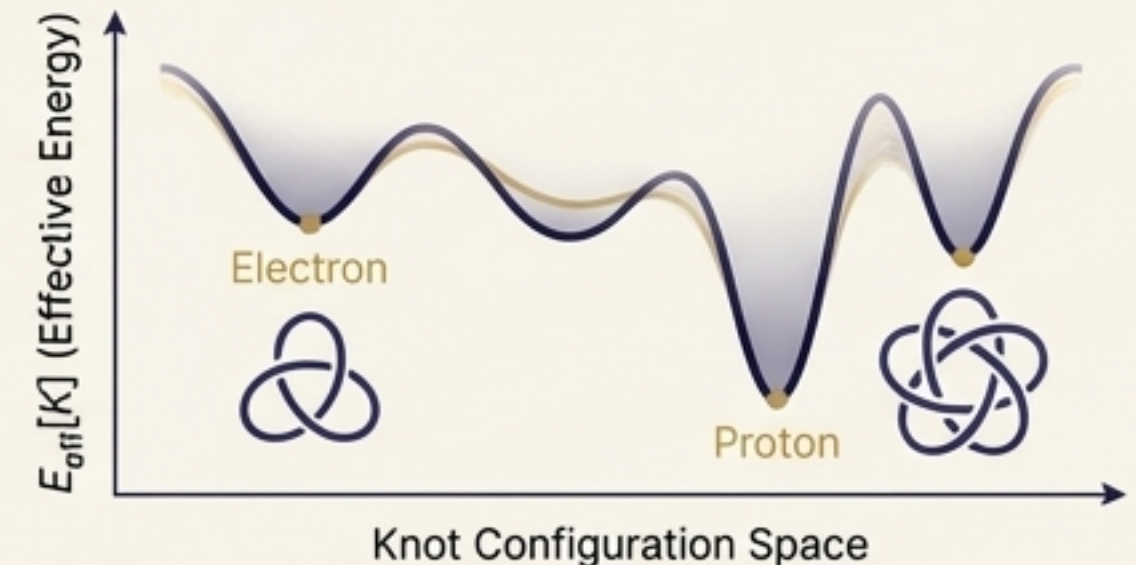
- This implies testable, direction-dependent effects in neutrino propagation, correlated with gradients in the clock/gravitational field.



Mass Spectrum

Discrete Mass Spectrum:

- The stability of particles is governed by an effective energy functional $E_{\text{eff}}[K]$.
- Numerical minimization of this functional for different knot classes should reproduce the observed particle mass ratios.

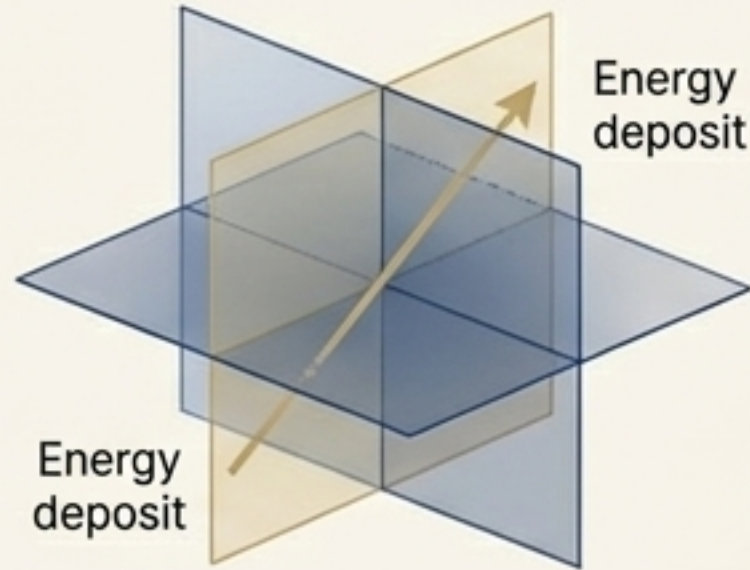


The Path Forward: A Living Research Program

SST unifies time, gravity, and mass within a single hydrodynamic framework, anchored by the falsifiable prediction of relational time-of-arrival a prediction of relational time-of-arrival.

Immediate Next Steps

The framework is well-suited for a focused program of theoretical and numerical study:



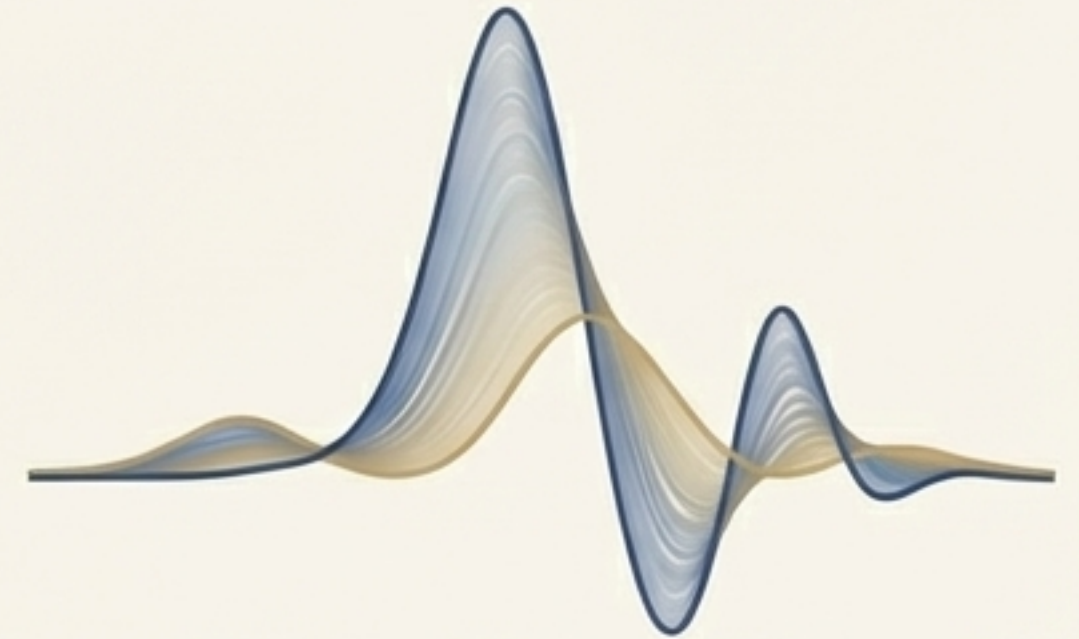
3+1D TOA Calculations

Extend the worked example to finite-size detectors to quantify geometric dependence.



Numerical Knot Energetics

Model the effective energy functional $E_{\text{eff}}[K]$ to derive the mass spectrum from first principles and test the stability of different knot configurations.



Non-Gaussian Clock Dynamics

Explore clock sectors beyond the simple quadratic EFT to predict higher-cumulant signatures and non-Gaussian features in the TOA distribution $p(\Theta)$.